

To: Prof. Luisa Verdoliva, Editor-in-Chief, IEEE Transactions on Information Forensics and Security (T-IFS)

Subject: Formal Resubmission & Point-by-Point Response to Prescreening Decision for Manuscript #T-IFS-28992-2026

Title: "Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering Under Extreme Imbalance and Topological Camouflage"

Dear Prof. Luisa Verdoliva and Members of the Editorial Board,

Thank you for the constructive prescreening evaluation of our manuscript #T-IFS-28992-2026. We deeply value the editorial board's commitment to the highest standards of presentation, clarity, and technical depth. In accordance with IEEE Signal Processing Society (SPS) Policy 6.16, we submit this substantially revised and visually overhauled manuscript along with a complete itemized response to every concern raised during prescreening.

Summary of Major Revisions in Response to Prescreening Feedback

1. Section 3 Methodology Clarity & Unified Algorithmic Specification (Editor Concern 1):

- Pedagogical Structural Reorganization: Section 3 has been completely restructured into a modular, pedagogical sequence. Each architectural component (Taint & Invariants, Tri-Band Continuous Time Attention, Edge-Trust Gating, Typology GraphSMOTE, Evidence-Adaptive Fusion, and Conformal Risk Control) is now prefaced by an intuitive domain motivation paragraph explaining the exact forensic mechanism before formal mathematics.
- Addition of Unified Algorithm 1: We added formal pseudocode (Algorithm 1: C-STGB Unified Multi-Stage Pipeline) detailing all five phases (feature extraction, edge pruning/oversampling, temporal aggregation, risk-sensitive calibration, and streaming 3-tier triage) with explicit systems time complexities ($O(K \cdot T_{\text{iter}})$) and verified streaming SLAs (0.45 ms per event).
- Equation Formatting: Split all multiline formulations across display math to eliminate overfull boxes and enhance readability.

2. Visual Legibility of Figures and Tables (Editor Concern 2):

- Table Redesign & Un-Shrunk Typography: Eliminated all aggressive table squishing. Table 2 (14-dataset benchmark scorecard) was redesigned into a spacious full-width table spanning the full page width with balanced row height, legible typography, and zero horizontal clipping. Tables 1, 3, 5, and 6 were reformatted with clean column layouts, balanced row spacing, and clear standard font sizes.
- Figure Overhaul & Vector Legibility: All figures were overhauled as high-resolution native vector graphics with legible typography: Figure 1 (System Architecture) was regenerated as an ultra-crisp vector diagram spanning the full page width ($7.16'' \times 2.35''$); Figure 2 (PR-ROC and t-SNE) features high-contrast axes and clear labels; Figure 3 (Camouflage Robustness) is rendered as a clean single-column vector graphic with high-contrast legends; and Figure 4 (15-Node Bitcoin Subgraph Case Study) clearly displays every transaction flow arrow, structuring burst, and wash loop.

3. Comprehensive State-of-the-Art Comparison & Recent TIFS Literature (Editor Concern 3):

- Integration of Flagship TIFS / IEEE Literature: We updated Section 2 (Related Work) and benchmark discussions to explicitly contextualize and differentiate C-STGB against flagship recent works:
 - Luo et al., "MLaD²: A Semi-Supervised Money Laundering Detection Framework Based on Decoupling Training," IEEE TIFS, vol. 19, 2024 (decoupled semi-supervised learning vs. our risk-controlled temporal conformal framework).
 - Ou et al., "CamFD: Semi-Supervised Camouflage-Aware Fraud Detection Based on Dynamic Graphs," IEEE TSMC, vol. 56, 2026 (dynamic camouflage filtering vs. our learnable edge-trust gating and degree safety valves).
 - Cheng et al., "Anti-Money Laundering by Group-Aware Deep Graph Learning," IEEE TKDE, vol. 35, 2023 (group-level aggregation vs. our individual class-conditional conformal containment).
- Scientific Positioning: We clarify empirical domain boundaries: C-STGB achieves statistically significant superiority over deep GNN baselines ($W = 105.0, p_{\text{adj}} < 0.001$), parity with tuned gradient-boosted trees globally, and marked superiority on complex relational multi-hop topologies (+9.98 pp on IBM-AMLSim HI).

4. Strict Space Budget & IEEE Policy Compliance:

- The main manuscript complies strictly with the IEEE SPS page ceiling at exactly 13.0 pages (inclusive of all 53 references and author biographies).
- The Supplementary Material is strictly 5.0 pages (within the ≤ 6.0 -page limit), providing complete mathematical proofs, 14-dataset itemizations, and hyperparameter grids.
- All code, reproducible seeds, and pipeline artifacts are publicly archived: <https://github.com/NazmulHasanNihal/Intelligent-AML>.

We confirm that this manuscript represents original research, has not been published elsewhere, and is not under consideration by another journal. All co-authors have reviewed and approved this revised version. We look forward to the manuscript progressing to full peer review.

Sincerely yours,

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